

Commerce XI Maths App

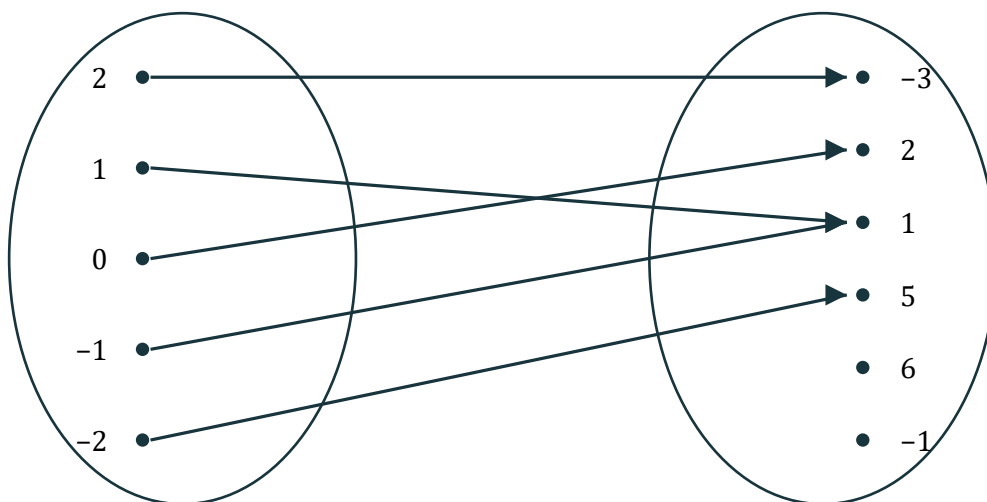
Exercise 2.1

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 2 - Functions | Total questions: 31

Question 1 - Q1(a)

Check if the relation shown in the arrow diagram is a function.



Options

- A. It is a function
- B. It is not a function because 1 has two images
- C. It is not a function because -2 has no image
- D. It is not a function because two inputs have image 1

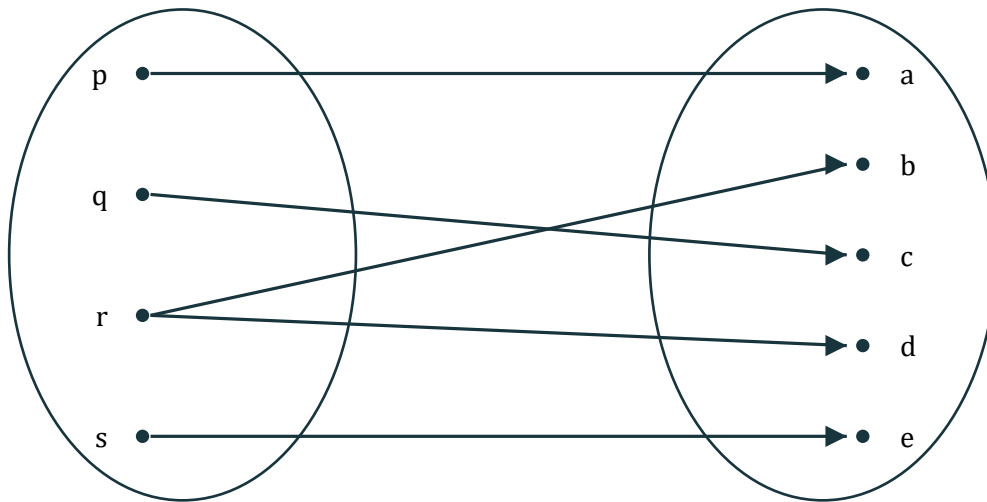
Correct answer: It is a function

Explanation and steps

A relation is a function only when every element in the left set has exactly one image. Inspecting every arrow gives the stated conclusion.

Question 2 - Q1(b)

Check if the relation shown in the arrow diagram is a function.



Options

- A. It is not a function because r has two images
- B. It is a function
- C. It is not a function because q has no image
- D. It is not a function because a has one pre-image

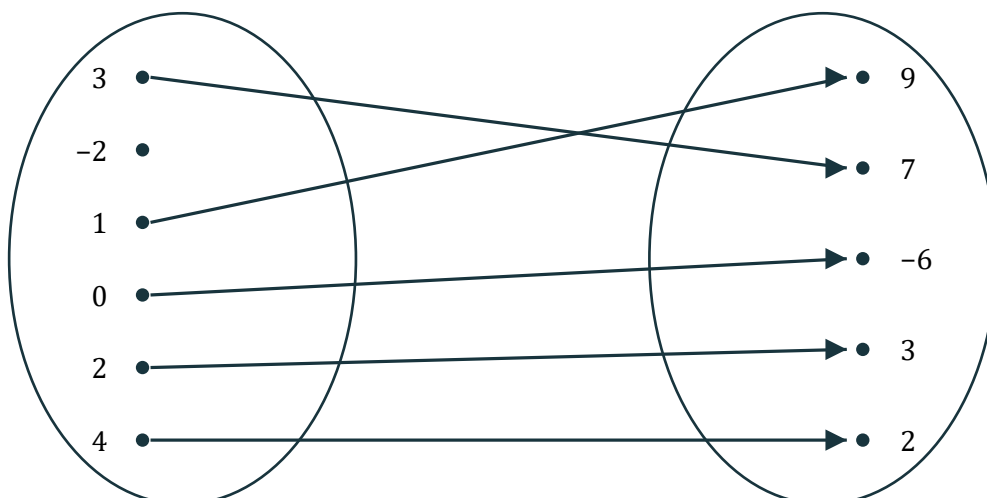
Correct answer: It is not a function because r has two images

Explanation and steps

A relation is a function only when every element in the left set has exactly one image. Inspecting every arrow gives the stated conclusion.

Question 3 - Q1(c)

Check if the relation shown in the arrow diagram is a function.



Options

- A. It is not a function because -2 has no image
- B. It is a function
- C. It is not a function because 3 and 1 have different images
- D. It is not a function because 0 has image -6

Correct answer: It is not a function because -2 has no image

Explanation and steps

A relation is a function only when every element in the left set has exactly one image. Inspecting every arrow gives the stated conclusion.

Question 4 - Q2(a)

Does $\{(1, 0), (3, 3), (2, -1), (4, 1), (2, 2)\}$ represent a function from $A = \{1, 2, 3, 4\}$ to $B = \{-1, 0, 1, 2, 3\}$?

Options

- A. No, because 2 has two images
- B. Yes, it is a function
- C. No, because 3 has no image
- D. No, because 0 is in B

Correct answer: No, because 2 has two images

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives No, because 2 has two images. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 5 - Q2(b)

Does $\{(1, 2), (2, -1), (3, 1), (4, 3)\}$ represent a function from $A = \{1, 2, 3, 4\}$ to $B = \{-1, 0, 1, 2, 3\}$?

Options

- A. Yes, it is a function
- B. No, because 0 has no pre-image
- C. No, because -1 is negative
- D. No, because all images must be distinct

Correct answer: Yes, it is a function

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives Yes, it is a function. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 6 - Q2(c)

Does $\{(1, 3), (4, 1), (2, 2)\}$ represent a function from $A = \{1, 2, 3, 4\}$ to $B = \{-1, 0, 1, 2, 3\}$?

Options

- A. No, because 3 in A has no image
- B. Yes, it is a function
- C. No, because 3 is an image
- D. No, because 4 cannot be a pre-image

Correct answer: No, because 3 in A has no image

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives No, because 3 in A has no image. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 7 - Q2(d)

Does $\{(1, 1), (2, 1), (3, 1), (4, 1)\}$ represent a function from $A = \{1, 2, 3, 4\}$ to $B = \{-1, 0, 1, 2, 3\}$?

Options

- A. Yes, it is a function
- B. No, because several elements have the same image
- C. No, because 1 appears four times
- D. No, because 0 has no pre-image

Correct answer: Yes, it is a function

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives Yes, it is a function. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 8 - Q3(a)

If $f(m) = m^2 - 3m + 1$, find $f(0)$.

Correct answer: 1

Explanation and steps

Substitute $m = 0$: $f(0) = 0^2 - 3(0) + 1 = 1$.

Question 9 - Q3(b)

If $f(m) = m^2 - 3m + 1$, find $f(-3)$.

Correct answer: 19

Explanation and steps

Substitute $m = -3$: $f(-3) = (-3)^2 - 3(-3) + 1 = 9 + 9 + 1 = 19$.

Question 10 - Q3(c)

If $f(m) = m^2 - 3m + 1$, find $f(1/2)$.

Options

- A. $-1/4$
- B. $1/4$
- C. $3/4$
- D. $-3/4$

Correct answer: $-1/4$

Explanation and steps

Substitute $m = 1/2$: $f(1/2) = 1/4 - 3/2 + 1 = (1 - 6 + 4)/4 = -1/4$.

Question 11 - Q3(d)

If $f(m) = m^2 - 3m + 1$, find $f(x + 1)$.

Options

- A. $x^2 - x - 1$
- B. $x^2 + x - 1$
- C. $x^2 - 3x + 1$
- D. $x^2 - x + 1$

Correct answer: $x^2 - x - 1$

Explanation and steps

Replace m by $x + 1$ and simplify: $(x + 1)^2 - 3(x + 1) + 1 = x^2 - x - 1$.

Question 12 - Q3(e)

If $f(m) = m^2 - 3m + 1$, find $f(-x)$.

Options

- A. $x^2 + 3x + 1$
- B. $x^2 - 3x + 1$
- C. $-x^2 + 3x + 1$
- D. $x^2 + 3x - 1$

Correct answer: $x^2 + 3x + 1$

Explanation and steps

Replace m by $-x$: $(-x)^2 - 3(-x) + 1 = x^2 + 3x + 1$.

Question 13 - Q4(a)

Find x if $g(x) = 0$, where $g(x) = (5x - 6)/7$.

Options

- A. $x = 6/5$
- B. $x = 5/6$
- C. $x = -6/5$
- D. $x = 7/5$

Correct answer: $x = 6/5$

Explanation and steps

Set the numerator equal to zero: $5x - 6 = 0$, hence $x = 6/5$.

Question 14 - Q4(b)

Find x if $g(x) = 0$, where $g(x) = (18 - 2x^2)/7$.

Options

- A. $x = \pm 3$
- B. $x = 3$ only
- C. $x = \pm 9$
- D. $x = \pm\sqrt{7}$

Correct answer: $x = \pm 3$

Explanation and steps

$18 - 2x^2 = 0$ gives $x^2 = 9$, hence $x = \pm 3$.

Question 15 - Q4(c)

Find x if $g(x) = 0$, where $g(x) = 6x^2 + x - 2$.

Options

- A. $x = 1/2$ or $-2/3$
- B. $x = -1/2$ or $2/3$
- C. $x = 2$ or -3
- D. $x = 1/3$ or -1

Correct answer: $x = 1/2$ or $-2/3$

Explanation and steps

Factor $6x^2 + x - 2 = (3x + 2)(2x - 1)$. Therefore $x = -2/3$ or $1/2$.

Question 16 - Q5

Find x if $f(x) = g(x)$, where $f(x) = x^4 + 2x^2$ and $g(x) = 11x^2$.

Options

- A. $x = 0, \pm 3$
- B. $x = \pm 3$ only
- C. $x = 0, \pm\sqrt{11}$
- D. $x = 0, \pm 9$

Correct answer: $x = 0, \pm 3$

Explanation and steps

Replace x by $x - 1$: $3(x - 1)^2 - 5(x - 1) + 7 = 3x^2 - 11x + 15$.

Question 17 - Q6(a)

If $f(x) = \begin{cases} x^2 + 3, & x \leq 2 \\ 5x + 7, & x > 2 \end{cases}$, then find $f(3)$

Correct answer: 22

Explanation and steps

Since $3 > 2$, use $5x + 7$: $f(3) = 15 + 7 = 22$.

Question 18 - Q6(b)

If $f(x) = \begin{cases} x^2 + 3, & x \leq 2 \\ 5x + 7, & x > 2 \end{cases}$, then find $f(2)$

Correct answer: 7

Explanation and steps

Since $2 \leq 2$, use $x^2 + 3$: $f(2) = 4 + 3 = 7$.

Question 19 - Q6(c)

If $f(x) = \begin{cases} x^2 + 3, & x \leq 2 \\ 5x + 7, & x > 2 \end{cases}$, then find $f(0)$

Correct answer: 3

Explanation and steps

Since $0 \leq 2$, use $x^2 + 3$: $f(0) = 3$.

Question 20 - Q7(a)

If $f(x) = \begin{cases} 4x - 2, & x \leq -3 \\ 5, & -3 < x < 3 \\ x^2, & x \geq 3 \end{cases}$, then find $f(-4)$

Correct answer: -18

Explanation and steps

Since $-4 \leq -3$, use $4x - 2$: $f(-4) = -16 - 2 = -18$.

Question 21 - Q7(b)

If $f(x) = \begin{cases} 4x - 2, & x \leq -3 \\ 5, & -3 < x < 3 \\ x^2, & x \geq 3 \end{cases}$, then find $f(-3)$

Correct answer: -14

Explanation and steps

Since $-3 \leq -3$, use $4x - 2$: $f(-3) = -12 - 2 = -14$.

Question 22 - Q7(c)

If $f(x) = \begin{cases} 4x - 2, & x \leq -3 \\ 5, & -3 < x < 3 \\ x^2, & x \geq 3 \end{cases}$, then find $f(1)$

Correct answer: 5

Explanation and steps

Since $-3 < 1 < 3$, use the middle rule: $f(1) = 5$.

Question 23 - Q7(d)

If $f(x) = \begin{cases} 4x - 2, & x \leq -3 \\ 5, & -3 < x < 3 \\ x^2, & x \geq 3 \end{cases}$, then find $f(5)$

Correct answer: 25

Explanation and steps

Since $5 \geq 3$, use x^2 : $f(5) = 25$.

Question 24 - Q8(a)

If $f(x) = 3x + 5$ and $g(x) = 6x - 1$, find $(f + g)(x)$.

Options

A. $9x + 4$

B. $9x + 6$

C. $3x + 4$

D. $18x^2 + 27x - 5$

Correct answer: $9x + 4$

Explanation and steps

Add the functions: $(3x + 5) + (6x - 1) = 9x + 4$.

Question 25 - Q8(b)

If $f(x) = 3x + 5$ and $g(x) = 6x - 1$, find $(f - g)(2)$.

Correct answer: 0

Explanation and steps

$f(2) = 11$ and $g(2) = 11$, so $(f - g)(2) = 0$.

Question 26 - Q8(c)

If $f(x) = 3x + 5$ and $g(x) = 6x - 1$, find $(fg)(3)$.

Correct answer: 238

Explanation and steps

$f(3) = 14$ and $g(3) = 17$, so $(fg)(3) = 14 \times 17 = 238$.

Question 27 - Q8(d)

If $f(x) = 3x + 5$ and $g(x) = 6x - 1$, find $(f/g)(x)$ and its domain.

Options

A. $(3x + 5)/(6x - 1)$, domain $\mathbb{R} - \{1/6\}$

B. $(6x - 1)/(3x + 5)$, domain $\mathbb{R} - \{-5/3\}$

C. $(3x + 5)/(6x - 1)$, domain $\mathbb{R}$

D. $9x + 4$, domain $\mathbb{R}$

Correct answer: $(3x + 5)/(6x - 1)$, domain $\mathbb{R} - \{1/6\}$

Explanation and steps

$(f/g)(x) = (3x + 5)/(6x - 1)$. Exclude $6x - 1 = 0$, so $x \neq 1/6$.

Question 28 - Q9(a)

If $f(x) = 2x^2 + 3$ and $g(x) = 5x - 2$, then find $f \circ g$

Options

A. $50x^2 - 40x + 11$

B. $10x^2 + 13$

C. $25x^2 - 20x + 7$

D. $50x^2 - 20x + 5$

Correct answer: $50x^2 - 40x + 11$

Explanation and steps

$$f[g(x)] = 2(5x - 2)^2 + 3 = 50x^2 - 40x + 11.$$

Question 29 - Q9(b)

If $f(x) = 2x^2 + 3$ and $g(x) = 5x - 2$, then find $g \circ f$

Options

A. $10x^2 + 13$

B. $50x^2 - 40x + 11$

C. $10x^2 + 15$

D. $25x^2 - 4$

Correct answer: $10x^2 + 13$

Explanation and steps

$$g[f(x)] = 5(2x^2 + 3) - 2 = 10x^2 + 13.$$

Question 30 - Q9(c)

If $f(x) = 2x^2 + 3$ and $g(x) = 5x - 2$, then find $f \circ f$

Options

A. $8x^4 + 24x^2 + 21$

B. $4x^4 + 12x^2 + 12$

C. $8x^4 + 12x^2 + 21$

D. $4x^4 + 24x^2 + 9$

Correct answer: $8x^4 + 24x^2 + 21$

Explanation and steps

$$f[f(x)] = 2(2x^2 + 3)^2 + 3 = 8x^4 + 24x^2 + 21.$$

Question 31 - Q9(d)

If $f(x) = 2x^2 + 3$ and $g(x) = 5x - 2$, then find $g \circ g$

Options

A. $25x - 12$

B. $25x - 4$

C. $10x - 4$

D. $25x + 12$

Correct answer: $25x - 12$

Explanation and steps

$$g[g(x)] = 5(5x - 2) - 2 = 25x - 12.$$